

mauttos-randtools

A library for generating random numerical lists, developed with a focus on mathematical applications, logic testing, and programming education.

About the Project

The `marcos_randtools` library provides a simple interface to encapsulate randomization logic, structural validations, and loops. The goal is to provide a lightweight tool for generating random datasets while maintaining the mathematical integrity of operations, eliminating the need to rewrite validation functions in every new project.

How to Use (Parameters and Examples)

The main function of the library is `randlist`, which accepts up to four parameters to customize your list of numbers:

size (int)

The amount of numbers you want to generate in the list.

max_val (int or float)

The maximum value a generated number can reach (the generation always occurs from 1 to `max_val`).

allow_duplicates (bool, optional, default=False)

If set to `True`, allows numbers to repeat in the list. If `False`, ensures total exclusivity (all numbers will be different).

use_floats (bool, optional, default=False)

If set to `True`, generates real numbers (with decimal places). If `False`, generates only integers.

Practical Examples

Below, see 4 usage scenarios, ranging from the most basic to mathematical exception handling:

1. Basic Usage (Unique Integers)

```
from marcos_randtools import randlist

# Generating 6 unique numbers between 1 and 60
draw = randlist(size=6, max_val=60) # or draw = randlist(6, 60)
print(draw)

# Possible output: [12, 45, 7, 58, 23, 31]
```

2. Allowing Duplicates

```
from marcos_randtools import randlist

# Simulating 10 rolls of a 6-sided die
rolls = randlist(size=10, max_val=6, allow_duplicates=True) # or rolls = randlist(10, 6, True)
print(rolls)

# Possible output: [3, 6, 1, 3, 2, 6, 5, 1, 4, 3]
```

3. Advanced Level (Generating Real Numbers)

To generate decimal numbers (floats) without repetition, we pass `False` as the third parameter and `True` as the fourth.

```
from marcos_randtools import randlist

# Generating 3 unique real values between 1 and 10
# (You could also use keyword arguments: randlist(size=3, max_val=10, use_floats=True))
real_values = randlist(3, 10, False, True)
print(real_values)

# Possible output: [7.34512, 1.98345, 4.56789]
```

4. Error Handling (Pigeonhole Principle)

The library features **Internal Mathematical Security**. The Pigeonhole Principle states that if you have N pigeons and M holes, and $N > M$, at least one hole must contain more than one pigeon.

The Mathematical Reality in Code

In programming, this means it is impossible to generate a list of exclusive integers if the desired quantity is greater than the available range. If this is attempted, the algorithm intercepts the failure and raises a `ValueError` to prevent infinite loops.

```
from marcos_randtools import randlist

# Trying to generate 10 UNIQUE numbers between 1 and 5 (Mathematically impossible)
try:
    math_error = randlist(10, 5)
except ValueError as error:
    print("Captured error:", error)

# Guaranteed output:
# Captured error: Impossible to generate 10 unique integers within a limit of 5.
```

Installation

To install the library in your development environment, use the standard Python package manager:

```
pip install mauttos-randtools
```